HISTORICAL CLIFFORD HESSIAN IDENTITIES

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Historical Clifford Hessian formulas

Let $(A_0, \dots, A_q)$ be a symmetric Clifford system on $\mathbb{R}^{2m}$, with $A_iA_j + A_jA_i = 2\delta_{ij}I$. For $(y, z) \in \mathbb{R}^{2m} \oplus \mathbb{R}^{q+1}$ put

$$\tau_i = \langle A_i y, y\rangle, \quad A_z = \sum_i z_i A_i, \quad A_\tau = \sum_i \tau_i A_i, \quad u = |y|^2, \quad v = |z|^2, \quad T = \sum_i \tau_i^2, \quad R = u + v,$$

$$F = \sum_i z_i\tau_i, \quad U = \nabla F = (2A_z y, \tau), \quad S = |U|^2 = 4uv + T, \quad Z = \nabla S = (8vy + 4A_\tau y, 8uz), \quad J = |Z|^2.$$

The original labels are retained to match the historical verification units; the gradient contraction is written as $J = |\nabla S|^2$ in the current notation. These formulas are not needed by the shorter proof in the revised Clifford chapter.

For tangent vectors $V = (\xi, \zeta)$ and $W = (\eta, \vartheta)$,

$$\operatorname{Hess} F(V, W) = 2\langle A_z\xi, \eta\rangle + 2\langle A_\zeta y, \eta\rangle + 2\langle A_\vartheta y, \xi\rangle, \tag{0.1}$$

$$\begin{aligned}\operatorname{Hess} S(V, W) = {} & 8v\langle\xi, \eta\rangle + 8\sum_{i=0}^{q}\langle A_i y, \xi\rangle\langle A_i y, \eta\rangle + 4\langle A_\tau\xi, \eta\rangle \\ & + 16\langle y, \xi\rangle\langle z, \vartheta\rangle + 16\langle y, \eta\rangle\langle z, \zeta\rangle + 8u\langle\zeta, \vartheta\rangle.\end{aligned} \tag{0.2}$$

Proposition 0.1 (Hessian identities)**.** *Let*

$$M = 4u^2 + 12uv + 4v^2 + T.$$

Then

$$\operatorname{Hess} F(U, U) = 8RF, \tag{0.3}$$

$$\operatorname{Hess} F(U, Z) = \frac{1}{2}J, \tag{0.4}$$

$$\operatorname{Hess} F(Z, Z) = 32FM, \tag{0.5}$$

$$\operatorname{Hess} S(U, U) = 32uv^2 + 32u^2v + 8uT - 16vT + 96F^2, \tag{0.6}$$

$$\operatorname{Hess} S(U, Z) = F(128u^2 + 640uv + 128v^2 + 96T), \tag{0.7}$$

$$\begin{aligned}\operatorname{Hess} S(Z, Z) = {} & 512uv^3 + 2048u^2v^2 + 512u^3v \\ & + 1280v^2T + 1920uvT + 128u^2T + 64T^2.\end{aligned} \tag{0.8}$$

1. Derivation of the Hessian identities

We derive the identities in Proposition 0.1 term by term. Write

$$B = A_z, \qquad C = A_\tau.$$

For $V = (\xi, \zeta)$, expansion in one variable gives

$$\begin{aligned}F(y + t\xi, z + t\zeta) = {} & F(y, z) + t\big(2\langle By, \xi\rangle + \langle A_\zeta y, y\rangle\big) \\ & + t^2\big(\langle B\xi, \xi\rangle + 2\langle A_\zeta y, \xi\rangle\big) + t^3\langle A_\zeta\xi, \xi\rangle.\end{aligned}$$

This gives (0.1). Since $S = 4uv + T$ and $T = \sum_i \tau_i^2$, the product rule gives (0.2).

Direct differentiation gives

$$U = (2By, \tau), \qquad Z = (8vy + 4Cy, 8uz).$$

Using $B^2 = v\,\mathrm{Id}$, $C^2 = T\,\mathrm{Id}$, and $BC + CB = 2F\,\mathrm{Id}$ in (0.1) gives

$$\operatorname{Hess} F(U,U) = 8RF, \qquad \operatorname{Hess} F(U,Z) = \frac{1}{2}J.$$

If $D = 2v\,\mathrm{Id} + C$, then the y-component of Z is $4Dy$, and

$$y^T DBDy = F(4v^2 + 4uv + T), \qquad y^T BDy = F(u + 2v).$$

Substitution into (0.1) gives $\operatorname{Hess} F(Z,Z) = 32FM$.

To evaluate $\operatorname{Hess} S$ on U and Z, put

$$p_i(V) = \langle A_i y, V_y \rangle.$$

The elementary quantities are

$$p_i(U) = 2uz_i, \qquad p_i(Z) = 4(u + 2v)\tau_i, \tag{1.1}$$

$$\langle y, U_y \rangle = 2F, \qquad \langle z, U_z \rangle = F, \tag{1.2}$$

$$\langle y, Z_y \rangle = 8uv + 4T, \qquad \langle z, Z_z \rangle = 8uv. \tag{1.3}$$

For $\operatorname{Hess} S(U,Z)$, the six terms of (0.2) are

1	$64uvF + 128v^2F$
2	$64u^2F + 128uvF$
3	$64uvF + 32TF$
4	$256uvF$
5	$128uvF + 64TF$
6	$64u^2F$

and their sum is (0.7).

For $\operatorname{Hess} S(Z,Z)$, use

$$\begin{aligned} |Z_y|^2 &= 64uv^2 + 64vT + 16uT, \\ \sum_i p_i(Z)^2 &= 16(u + 2v)^2 T, \\ \langle CZ_y, Z_y \rangle &= 64v^2T + 64uvT + 16T^2, \\ |Z_z|^2 &= 64u^2v. \end{aligned}$$

The five combined contributions are

1	$512uv^3 + 512v^2T + 128uvT$
2	$128u^2T + 512uvT + 512v^2T$
3	$256v^2T + 256uvT + 64T^2$
4	$2048u^2v^2 + 1024uvT$
5	$512u^3v$

and sum to (0.8). For (0.6), substitute U into (0.2) and use $BCB = 2FB - vC$. This gives

$$\operatorname{Hess} S(U,U) = 32uv^2 + 32u^2v + 8uT - 16vT + 96F^2.$$